Randomized and low-rank approximation

Sharp sparsity threshold for injectivity of a random sparse rectangular matrix

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Challenging because a sharp lower-singular-value threshold needs both necessity and sufficiency; community impact is the sparsity cost of reliable randomized embeddings.

Problem statement

Let $k \to \infty$ through multiples of 100, $n_k \geq k^{1+c}$ for fixed $c > 0$, and $1 \leq s_k \leq k$. Independently for each column of $M_k \in \mathbb{R}^{k \times n_k}$, choose s_k distinct row locations uniformly and put independent signs $\pm 1/\sqrt{s_k}$ there. Independently select a uniform $k/100$-element column set I_k.

Determine necessary and sufficient asymptotic conditions on s_k for the existence of an absolute $a > 0$ such that

$$\Pr\{\sigma_{\min}((M_k)_{:I_k}) \geq a\} = 1 - o(1).$$

Probability includes both random constructions. The requested threshold concerns a lower singular-value bound, not an upper distortion estimate.

Precision of the target

A complete resolution should give an explicit asymptotic criterion on the sparsity sequence (s_k) that is equivalent to the displayed probability bound for some fixed positive a. Separate necessary and sufficient estimates are partial progress whenever a gap between them leaves this property undecided. There is no blanket allowance to ignore constant or logarithmic factors: they need to be resolved insofar as they affect whether the property holds. Optimizing the numerical value of a is not required.

The paragraph following Problem 7.2 in the source suggests a natural intermediate goal: establish whether there is a critical logarithmic exponent $\alpha_* > 0$ and determine its value. The proposed separation is failure of the displayed property for $s_k = O((\log k)^{\alpha_* - \varepsilon})$ and success for $s_k = \Omega((\log k)^{\alpha_* + \varepsilon})$, for every fixed $\varepsilon > 0$ in the admissible sparsity range. Such a result need not settle the behavior at $s_k = (\log k)^{\alpha_* + o(1)}$, including possible constant or $\log \log k$ corrections. It is therefore a milestone toward the full target, rather than a replacement for it; existence of such an exponent is not assumed.

Reference

Huang, Rudelson, and Tikhomirov, *Well-Invertible Column Subsets of Sparse Matrices Are Rare*, §7, Problem 7.2.

Status check

Searches included *"Optimal OSI sparsity" "2026"* and the paper title with *threshold*. No sharp threshold result was located. A sufficient condition from a two-sided embedding theorem would not on its own establish necessity here.

Audit — 2026-09-10

Rechecked Problem 7.2 and Tikhomirov's later paper, which uses different aspect-ratio and randomness regimes. Threshold and sparse-rectangular-matrix searches found no sharp criterion for this fixed-column-sparsity model. A sufficient embedding result alone does not determine the target.

Clarification — 2026-09-12

Following issue #150, rechecked Problem 7.2 and its following paragraph and made the requested precision explicit. This is a clarification of the original target, not a new literature-wide status audit or a resolution. The ID, matrix model, probability bound and Open status are unchanged.